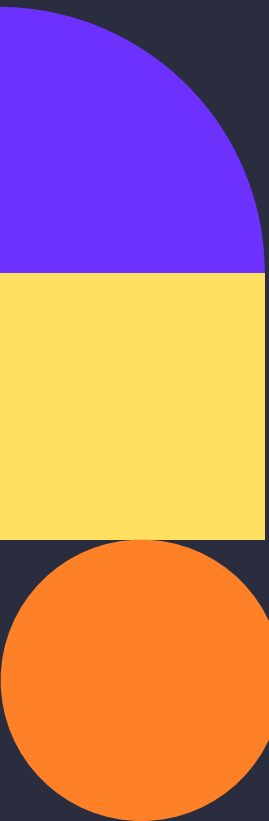


Information Gathering using Shodan CLI

Liam Salazar-Endara | 28/07/2025



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Introduction to SHODAN

SHODAN is a search engine for internet-connected devices. It allows cybersecurity professionals and researchers to gather intelligence on exposed services, devices, and potential vulnerabilities globally. In this project, SHODAN CLI was used to perform basic information gathering on one HTTP and one HTTPS target website to understand its usage in reconnaissance.



Tools and the OS

Requirements:

1. Operating System - Kali Linux (Installed in VirtualBox)
2. Tool - SHODAN Command Line Interface (CLI)

Prerequisites:

1. VirtualBox installed on host system
2. Kali Linux image configured
3. Internet connectivity for installing and using SHODAN



SHODAN



VirtualBox



Installation Steps for Home Lab – Kali Linux and SHODAN

Step 1 - Install Kali Linux ISO image

- We are going to need to go to the official website of Kali Linux OS which is as follows:
<https://www.kali.org/get-kali/>
- Then we are going to head to the option of "Installer Images" and head to the first recommended option "Installer" of Kali Linux which should take about 4GB of space
- This should lead to the installment of the Kali Linux ISO image within your Downloads folder.

Step 2 - Install VirtualBox

- We are then going to need to head to the official website of VirtualBox, which is as follows:
<https://www.virtualbox.org/>
- We are then going to need to determine the specs of our computer to determine the architecture
- After doing so we then are going to click on "Get Started Download" should it match the specs or go through release versions to determine the one which matches.

Step 3 - Installation of Kali Linux ISO Image into VirtualBox

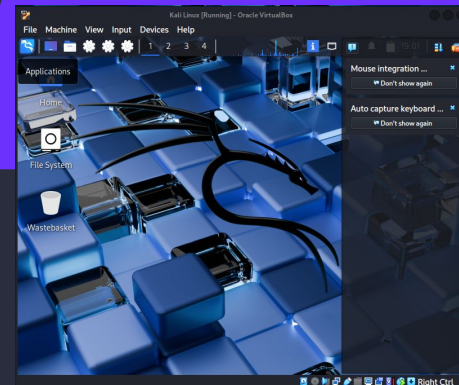
- Open **VirtualBox** → **New**.
- **Name:** Kali Linux
- **Type:** Linux | **Version:** Debian (64-bit) or Kali if listed.
- **RAM:** Minimum **2GB**, recommended **4GB+**.
- **Type:** VDI
- **Storage:** **Dynamically allocated**
- **Size:** **20GB+**
- Click **Create**, then **Start** to boot the VM and proceed with installation.
- Login with your **set credentials** after installation completes.

Step 4 - Kali Linux Installation Procedure

- We are then going to finally follow the installation procedure as outlined in the Kali Linux Documentation, which you can find here:
<https://www.kali.org/docs/installation/hard-disk-install/>
- Once in the Kali Desktop head to the terminal to follow with the installation of SHODAN

Step 5 - Installation of SHODAN on Kali Linux (Home Lab)

- Run the following in the terminal to install SHODAN starting with:
sudo apt-get update && upgrade -y
- 1. Install pipx (Python package installer)
sudo apt install pipx
- 2. Installation of shodan
pipx install shodan
- 3. Add the Setuptools
pipx inject shodan setuptools



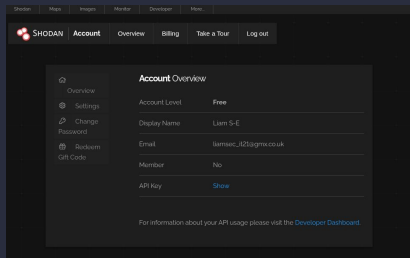
```
liam@LMD10:~$ pipx install shodan
installed package shodan 1.31.0, installed using Python 3.13.5
These apps are now globally available
- shodan
done! 🌟 🌟 🌟
liam@LMD10:~$ which shodan
/home/liam/.local/bin/shodan
liam@LMD10:~$
```

```
liam@LMD10:~$ pipx inject shodan setuptools
injected package setuptools into venv shodan
done! 🌟 🌟 🌟
```

Setting a Shodan Account

Step 1 - Creating a Shodan Account

- We are going to need to head to the official Shodan page and set up an account and verify it by going on the website below:
<https://account.shodan.io/register>
- After completion, when you login you should be presented with the Dashboard



- Under the Account Overview you can see the API key for which there are two:
 1. The REST API - allows us to search for specific data and get summaries
 2. The Streaming API - gives live feedback of data as SHODAN collects it, but you can't search this data
- Bear in mind that this depends on which version of SHODAN you are registering with whether paid or not

Basic Search Examples

1. apache - This will return servers which run Apache software
2. Microsoft-IIS/7.5 - This will return servers which run version 7.5 of Microsoft's Internet Information Services web server software
3. nginx - This will return servers which run Nginx, an open source web server
4. cisco-ios - Returns devices running Cisco's IOS OS and is used mostly by networking hardware like routers and switches
5. default password - This might show devices or services where the default passwords are still used

Filters

ip:	Filters results by a specific IP address.Ex: ip:192.168.2.1
asn:	Filters results by a specific ASN (Autonomous System Number) ID.Ex: asn:AS8160
hostname:	Filters results by a specific hostname or find values that match the hostname.Ex: hostname:google.com
port:	Filters results by a specific port number of a service or find particular ports that are open.Ex: port:21
net:	Filters results from a specified CIDR (Classless Inter-Domain Routing) block.Ex: net:192.0.2.0/24
isp:	Filters results by devices assigned a particular address (space) from a specified ISP (Internet Service Provider).Ex: isp:Bell
city:	Filters results by a specific city or find devices in a particular city.Ex: city:Vancouver
country:	Filters results by a specific two-digit country code or find devices in a particular country.Ex: country:CA
os:	Filters results by a particular operating system or search based on the operating system.Ex: os:Linux
product:	Filters results by a particular software or product identified in the banner.Ex: product:Apache
version:	Filters results by a specified software version.Ex: version:2.2.5
geo:	Search for specific GPS coordinates.Ex: geo:42.3601,-71.0589 (command line only)
before/after:	Find results within a specific timeframe.Ex: after:2022-01-01 (command line only)

Getting Started with SHODAN CLI

Step 1 - Initialise SHODAN CLI

- Before using any SHODAN commands, you need to authenticate with your API key.
`shodan init <API_Key>`
- This step is required only once per setup. It stores your API key locally so you can use SHODAN CLI commands without re-entering it.

```
liam@LMD10:~$ shodan init Dra4ELExxP3IswrEd5JjsR3TOfnRXXG
/home/liam/.local/share/pipx/venvs/shodan/lib/python3.13/site-packages/shodan/__main__.py:33: UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin to Setuptools<81.
  import pkg_resources
Successfully initialized
```

Step 2 - Verify SHODAN Installation (Optional)

- Use the command below to ensure SHODAN is correctly installed and the API key is set up.
shodan info
- This will show:
 - Available query credits
 - Number of scan credits
 - User account info

Step 3 - Get Help on Commands

- If you want to see all available SHODAN CLI commands:
shodan
- Or to get help for a specific command:
shodan <command> --help

SHODAN DORKING: Application of the SHODAN filters – city filter

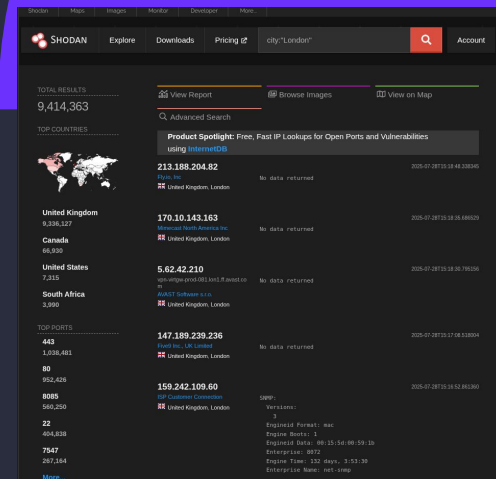
Using SHODAN for Targeted Searches (Dorking)

Performing a City-Based Search

- This step involves querying SHODAN for all internet-connected devices that are geographically located within a specific city
- In this example, we used the following filter:
City:"London"

Analysis of SHODAN Output

1. Search Query
 - The query used in the SHODAN search bar filters for all internet-connected devices geo located within London
2. Left Panel - Summary Statistics
 - Total Results - Displays the total number of devices identified by SHODAN within the specified location (London)
 - Top Countries - Although the search is city-specific, SHODAN also displays other countries that frequently appear in the dataset.
3. Commonly Detected Open Ports
 - These are the most frequently observed open ports on devices in the search results:
 - 443 - HTTPS (Secure Web Traffic)
 - 80 - HTTP (Unencrypted Web Traffic)
 - 8085 - Alternative HTTP Port
 - 22 - SSH (Remote Access/Login)
 - 7547 - CPE WAN Management Protocol (commonly used by ISPs for router management)
4. Device Information Listing
 - Each device listed in the output includes a range of metadata, such as:
 - IP Address
 - Organisation or Hosting Provider
 - Geolocation (City/Country)
 - Port and Service Information
 - Server Data (e.g., Engine ID, Enterprise Name)
 - Protocol Details (e.g. SNMP Version)

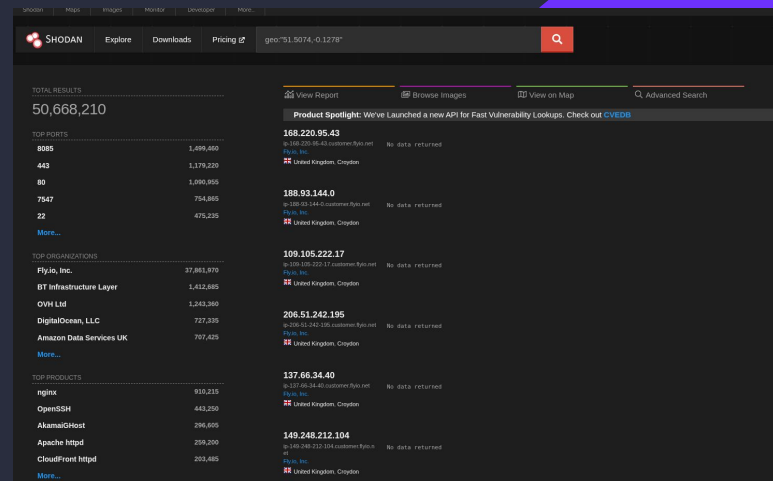


SHODAN DORKING: Application of the SHODAN filters – geo filter

Using SHODAN for Targeted Searches (Dorking)

Performing a Geolocation-Based Search

- The **geo** filter in SHODAN allows users to identify internet-connected devices based on specific latitude and longitude coordinates
- This is useful for pinpointing devices within a precise geographic area
- In this example, the following filter was used:
geo :“51.5074, -0.1278”
- This query would target devices located near the coordinates for central London



Analysis of SHODAN Output

- The results indicate that SHODAN identified approximately 50,668,210 devices associated with the specified geographic coordinates
- Additional statistical insights include:
 - Most commonly Used Parts - A breakdown of the ports that are most frequently open on the discovered devices #.
 - Top organizations - A list of the primary organisations or ISPs to which the devices belong.
 - Products and Services - Information about the technologies, applications, or services running on the devices
 - Public IP Addresses - A wide range of publicly accessible IP addresses associated with the devices in the specified location

SHODAN DORKING: Application of the SHODAN filters – country filter

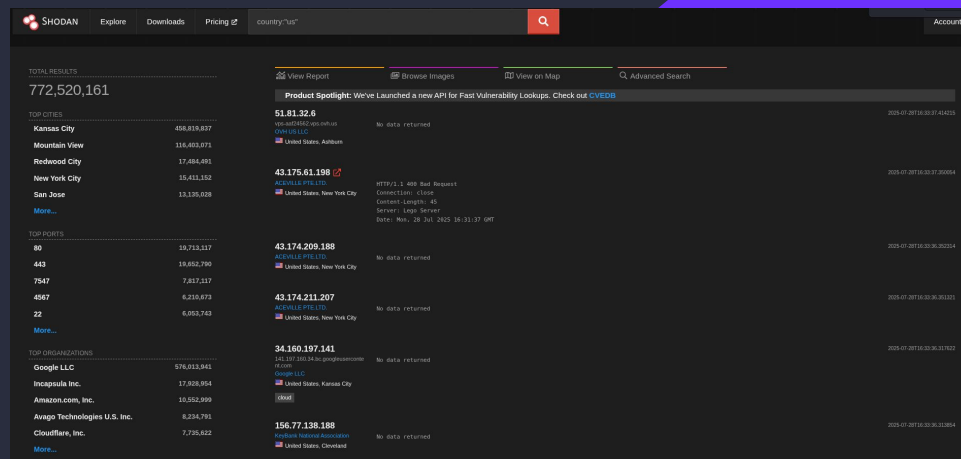
Using SHODAN for Targeted Searches (Dorking)

Performing a Country-Based Search

- The **country** filter in SHODAN enables users to identify internet-connected devices located within a specific country
- In this example, the following filter was used:
country:"us"
- This query would retrieve devices that are geographically located within the US

Analysis of SHODAN Output

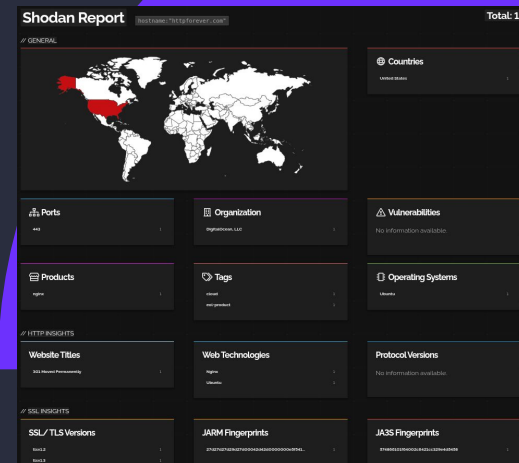
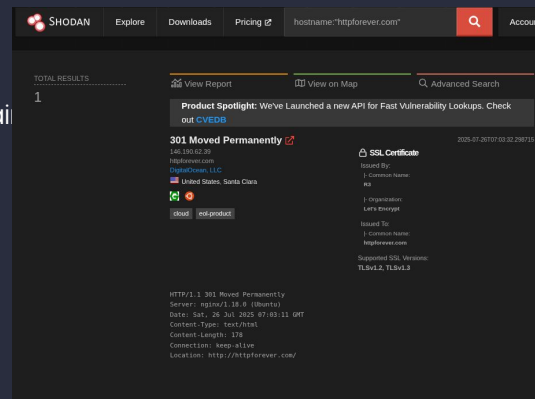
- The results from the country-based search provide several key insights:
 - Top Cities - The left panel displays the cities with the highest number of internet connected devices detected within the US where Kansas appears at the top, indicating the highest concentration of discovered devices
 - Most Commonly Used Ports - The output highlights the most frequently open ports across devices in the US. Notably, port 80 (HTTP) ranks highest, indicating widespread use of web services
 - Top Organisations - The statistics also reveal the leading organisation associated with the identified devices. Google LLC ranks at the top, reflecting a significant portion of devices communicating with or hosted by Google's infrastructure



SHODAN DORKING: Application of the SHODAN filters – hostname filter

Using SHODAN for Targeted Searches (Dorking) Performing a Hostname-Based Search

- The "hostname" filter in SHODAN allows users to locate internet-connected devices that match a specific domain or hostname
- In this example, we performed a search for devices associated with the following domain;
hostname:"httpforever.com"
- This query would return information about any device which are linked to or serve content under the specified hostname



Analysis of the SHODAN Output

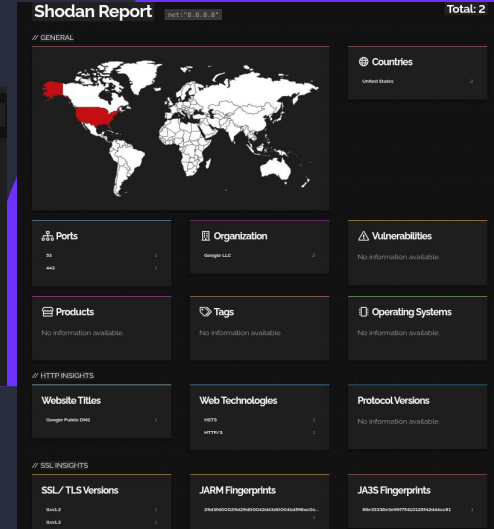
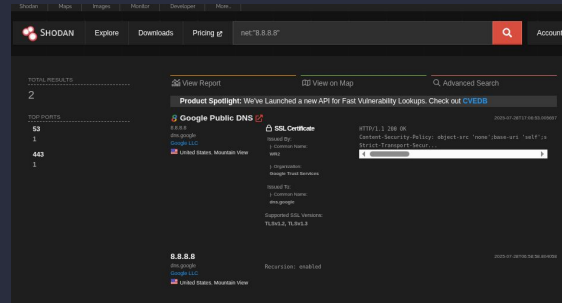
- The analysis indicates that a total of 1 device matches the hostname httpforever.com. The IP address associated with this device is 146.190.62.39
- The domain name is identified is also httpforever.com
- The geographic location of the host device is Santa Clara, United States
- The server is running Ngin, and the communication protocol in us is HTTP/1.1
- Additionally, the SSL certificate is issued by R3, a certificate authority operated by Let's Encrypt

SHODAN DORKING: Application of the SHODAN filters – net filter

Using SHODAN for Targeted Searching (Dorking)

Performing IP-Based Search

- The **net** filter in SHODAN enables users to locate internet-connected devices based on a specific IP address or a CIDR range
- In this example, we performed a search using the following query:
net:"8.8.8.8"
- This query would return some information about any device associated with the specified IP address



Analysis of SHODAN Output

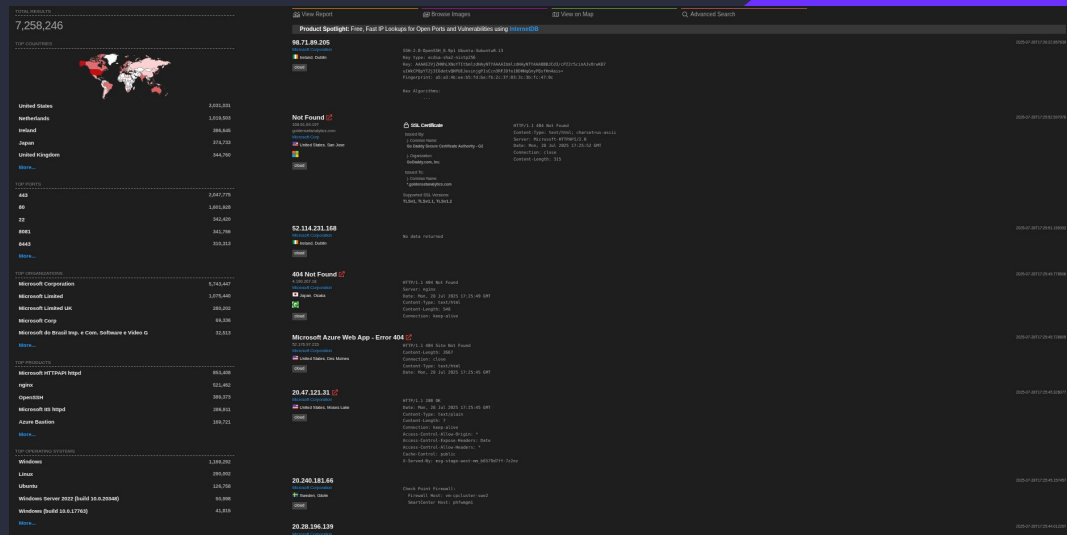
- The results from the IP-based search provide several key insights:
 - Total Devices Detected - The search revealed two devices associated with the IP address 8.8.8.8, which is publicly known as Google Public DNS
 - IP Ownership - The IP registered to Google LLC, confirming its affiliation with Google's global network infrastructure
 - SSL Certificate Issuer - The SSL certificate identified during the scan is issued by WR2, part of Google Trust Services, indicating secure communication channels managed by Google
 - Open Ports - The output highlights two open ports commonly associated with DNS and secure web services:
 - Port 53, used for Domain Name Systems queries
 - Port 443, used for HTTPS communication

SHODAN DORKING: Application of the SHODAN filters – org filter

Using SHODAN for Targeted Searches (Dorking)

Performing an Organisation-Based Search

- SHODAN provides the ability to search for internet-connected devices associated with a specific organisation using the **org** filter
- In this example, we executed the following query: **org:"Microsoft"**
- This search returns information about devices that are registered to or operated by Microsoft



Analysis of SHODAN Output

- The organisation-based search for "Microsoft" yields several notable insights:
 - Total Devices Identified – SHODAN reports a total of 7,258,246 internet-connected devices associated with Microsoft.
 - Geographic Distribution – The results indicate that Microsoft-operated infrastructure is present in multiple countries, with the United States ranking as the top country in terms of the number of detected devices linked to the organisation.
 - Commonly Used Ports – A wide range of ports are observed across the identified devices. Among them, port 443 (HTTPS) is the most prevalent, followed by ports 80 (HTTP), 22 (SSH), 8081, and others.
 - Operating Systems and Products – The most frequently detected operating system is Windows. The top organisation associated with these devices is Microsoft Corporation, and the most commonly observed product is Microsoft HTTPAPI httpd, indicating widespread deployment of Microsoft's proprietary web services.
-

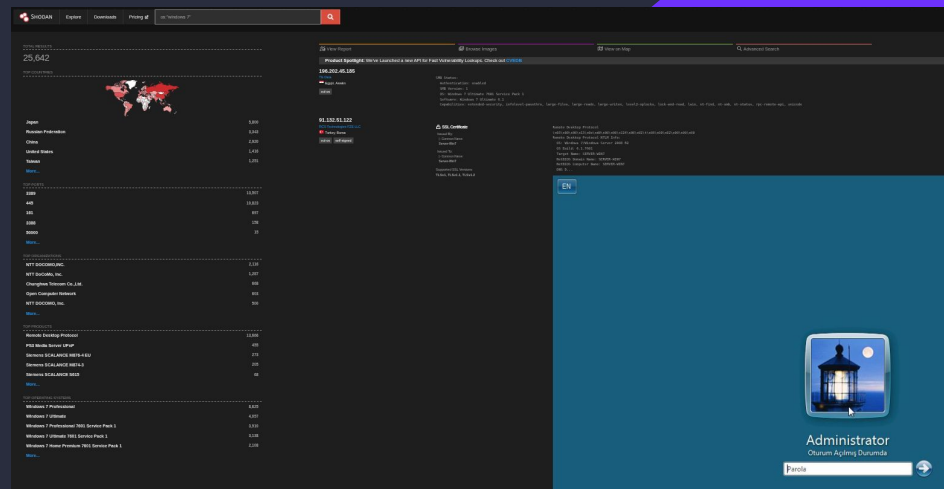
SHODAN DORKING: Application of the SHODAN filters – os filter

Using SHODAN for Targeted Searches (Dorking) Performing an Operating System-Based Search

- SHODAN allows users to search for internet-connected devices running a specific operating system using the **os** filter.
- In this example, the following query was used:

os:"Windows 7"

- This search returns information about devices that are running Windows 7, enabling the identification of legacy systems that may be exposed online



Analysis of SHODAN Output

- The operating system based search for "Windows 7" reveal several important findings:
 - Total Devices Detected – SHODAN identified a total of 25,642 internet-connected devices running the Windows 7 operating system.
 - Geographic Distribution – The majority of these devices are located in Japan, indicating a continued presence of Windows 7 systems in that region despite the end of official support.
 - Most Common Port – Port 3389, which is typically used for Remote Desktop Protocol (RDP), is the most frequently observed open port, suggesting that many of these systems may be remotely accessible.
 - Top Organisations – The leading organisation associated with these devices is NTT DOCOMO, INC., reflecting their significant use of Windows 7 infrastructure.
 - Commonly Used Products – The most prevalent service identified is Remote Desktop Protocol, indicating widespread use of remote access capabilities.
 - Server Message Block (SMB) – One of the devices in the listing reveals the presence of Server Message Block (SMB), a protocol used for file, printer, and resource sharing over a network, highlighting the potential exposure of internal services to the public internet.

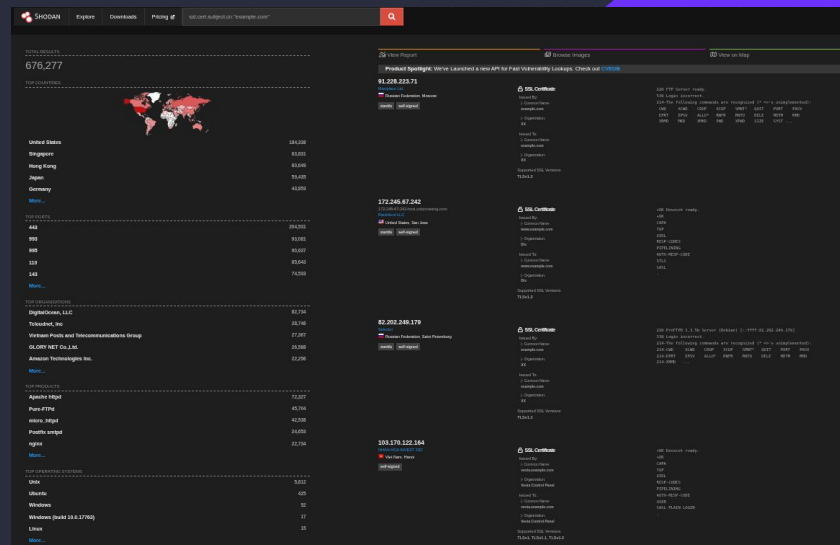
SHODAN DORKING: Application of the SHODAN filters – ssl/tls filter

Using SHODAN for Targeted Searches (Dorking) Performing an SSL/TLS Certificate-Based Search

- SHODAN provides the ability to search for devices or services that present an SSL/TLS certificate containing a specific Common Name (CN) in the certificate's subject field. This is useful for identifying hosts that serve content under a particular domain, even if the hostname or IP is not directly known.
- In this example, the following query was used:

ssl.cert.subject.cn:"example.com"

- This search returns results for devices or services whose SSL/TLS certificates include "example.com" as the Common Name, allowing for the discovery of related infrastructure based on certificate metadata.



Analysis of SHODAN Output

- The search results indicate that the SSL/TLS certificates presented include example.com as the Common Name.
- A total of 676,277 devices were identified presenting SSL/TLS certificates matching this criterion, with the majority of these devices located in the United States.
- The most commonly observed open port is 443, which is standard for HTTPS traffic.
- The leading organisation associated with these devices is DigitalOcean LLC.
- The most frequently detected product is Apache httpd, while the predominant operating system is Unix.
- Additionally, many of the certificates were identified as self-signed, as indicated by the presence of the "self signed" tag, which may have implications for certificate trustworthiness.

SHODAN DORKING: Application of the SHODAN filters – device filter

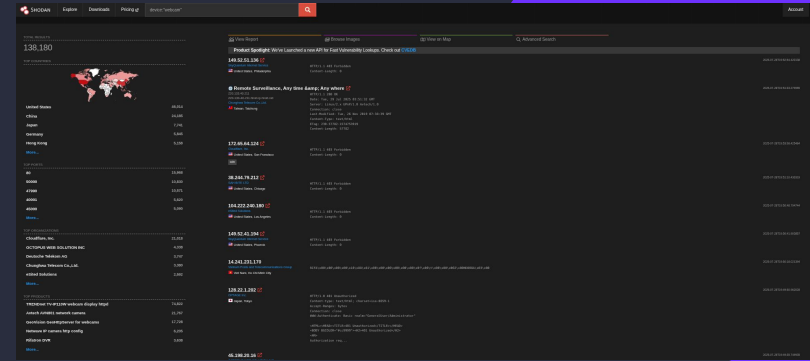
Using SHODAN for Targeted Searches (Dorking) Performing a Device Type Search

- SHODAN provides a search filter that enables users to locate specific types of internet-connected devices by specifying the device category.
- In this example, the query used was:

device:"webcam"

- This search returns results for devices identified as webcams that are publicly accessible on the internet.

The results of this query are illustrated in the accompanying image.



Analysis of SHODAN Output

- SHODAN identified a total of 138,180 devices classified as webcams.
- Among the devices found, one notable example is a surveillance system named “Remote Surveillance, Any time & Any Where”, located in Taiwan and owned by Chunghwa Telecom Co. Ltd.
- The device responds with a standard web server response, indicating that it is running on a Linux-based system within its software stack.
- The response content is a typical HTML web page, consistent with web interfaces commonly used for device management or monitoring.

SHODAN DORKING: Application of the SHODAN filters – http.html filter

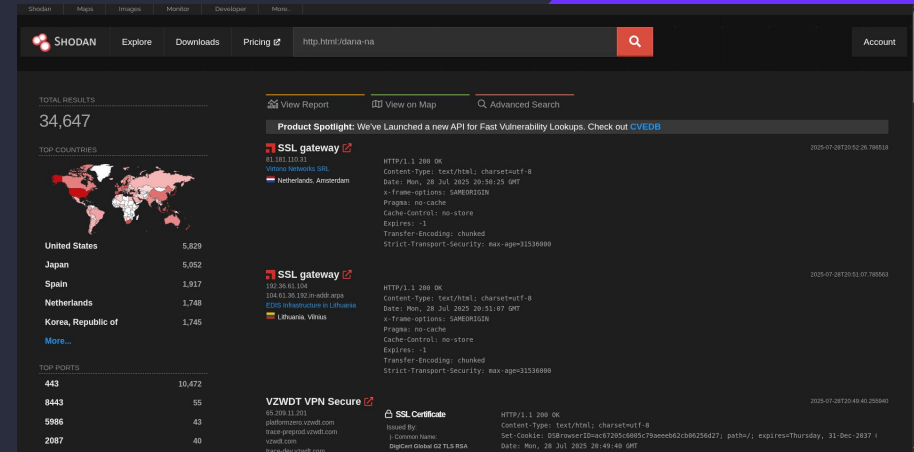
Using SHODAN for Targeted Searches (Dorking) Performing a Web Content Search

- This SHODAN dork is used to search for devices or servers that return an HTML page containing the specific string "/dana-na".
- The filter applied in this search is:

http.html:/dana-na

Analysis of SHODAN Output

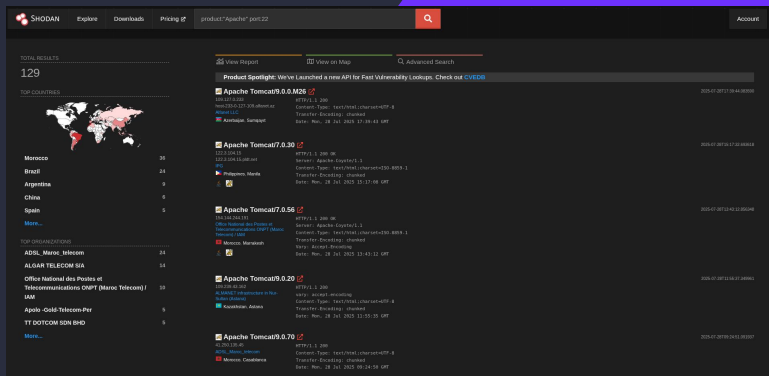
- The search identified a total of 34,647 devices worldwide with web interfaces containing the string "/dana-na".
- The country with the highest number of exposed devices is the United States.
- The most commonly observed open port among these devices is 443, which is typically used for secure HTTPS traffic.
- These devices are publicly accessible on the internet and may indicate the presence of systems running outdated Pulse Secure VPN firmware, potentially exposing them to security vulnerabilities.



SHODAN DORKING: Application of the SHODAN filters – product and port filters

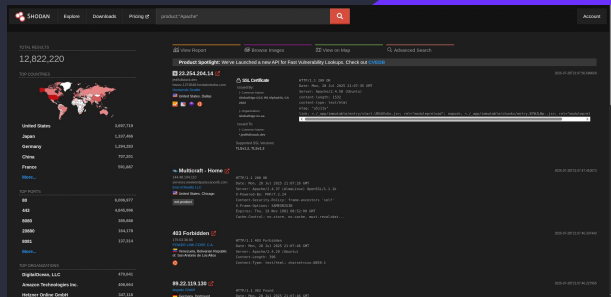
Using SHODAN for Targeted Searches (Dorking) Web Server Searching

- SHODAN's **product** filter allows users to search for devices running a specific software, service, or application.
- Additionally, the **product** filter enables searches for devices with a particular port number open and actively responding.
- In this example, we applied the following filters:
product:"Apache" port:"22"



Analysis of SHODAN Output

- The first image illustrates the results of a search filtered solely by devices running the Apache software.
- The second image demonstrates a more refined search, further restricting the results to devices with port 22 open.
- A noticeable reduction in the total number of devices is observed between the two searches, reflecting the added specificity of the port filter.
- The majority of responses returned are in the form of HTML web pages.
- Additionally, some servers transmit the response in chunked encoding, sending data in smaller segments rather than a single large block. This typically occurs when the server does not know the total size of the content at the start of the transmission.



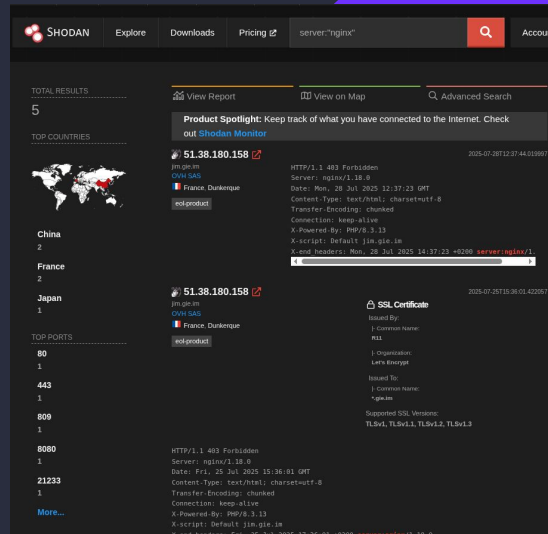
SHODAN DORKING: Application of the SHODAN filters – server filter

Using SHODAN for Targeted Searches (Dorking)

Performing a Server Header Search

- This search method allows users to identify devices where the HTTP response includes a “Server” header specifying a particular server software.
- In this example, we use the following query:

server:"nginx"



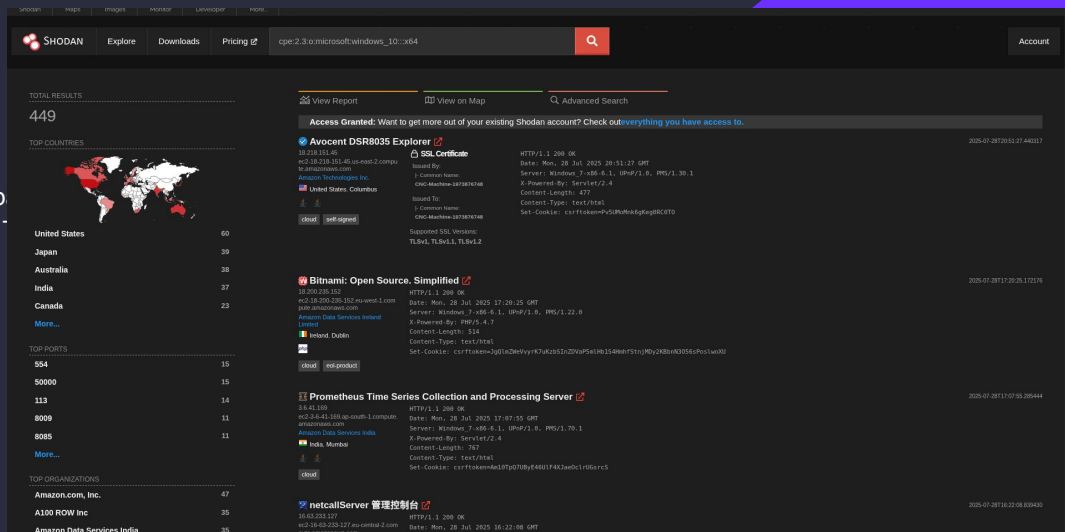
- The search identified a total of five devices with the HTTP “Server” header set to Nginx.
- Additional information available includes the devices’ public IP addresses, hostnames, Internet Service Providers (ISPs), and geographic locations.
- The presence of the X-Script header appears to be a custom HTTP header, indicating the specific internal script being executed on the server.
- Furthermore, the sites are running PHP version 8.3.13, suggesting that they serve dynamic web content with PHP enabled.
- The header “Connection: keep-alive” indicates that the server supports persistent connections, allowing multiple requests to be sent over the same connection without the need to re-establish it each time.

SHODAN DORKING: Application of the SHODAN filters – cpe filter

Using SHODAN for Targeted Searches (Dorking)

Step 1 – Performing a CPE (Common Platform Enumeration) Search

- The **cpe** filter in SHODAN enables the identification of devices based on specific software, operating systems, or hardware platforms. This feature is widely utilized in cybersecurity for vulnerability management and asset discovery.
- In this example, the following CPE query was used:
cpe:2.3:o:microsoft:windows_10:::x64



Analysis of SHODAN Output

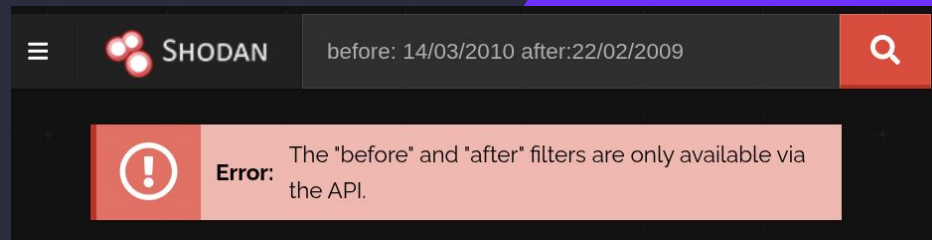
- The query targeted devices reporting their operating system as Microsoft Windows 10 (64-bit), based on CPE version 2.3 identifiers.
- SHODAN relies on banner grabbing, which requires devices to accurately report their software or operating system versions. However, devices may sometimes misreport or use custom headers (e.g., Server : Windows_7-x86-6.1), which can affect the accuracy of the classification.
- In such cases, SHODAN attempts to match the reported information to the closest known CPE. If even one data source—such as UPnP or an HTTP server header—indicates Windows 10, SHODAN may assign the device this CPE identifier, even if other banners suggest an earlier version like Windows 7.

SHODAN DORKING: Application of the SHODAN filters – before/after filter

Using SHODAN for Targeted Searches (Dorking)

Performing a Time-Range Search

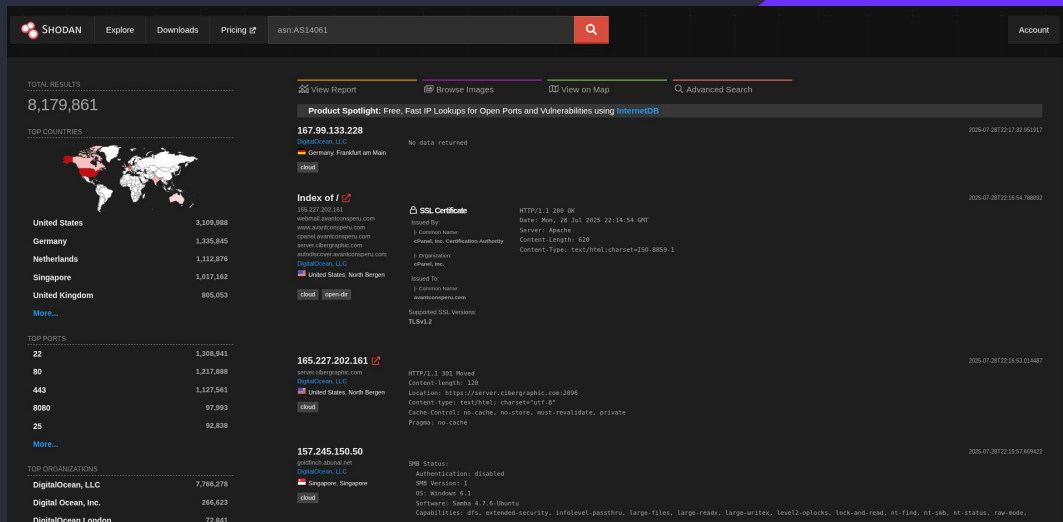
- SHODAN allows users to filter search results based on the time when devices were last detected, using the **before** and **after** filters to specify a date range.
- In this example, the following filters were applied:
after:22/02/2009 before:14/03/2010



SHODAN DORKING: Application of the SHODAN filters – asn filter

Using SHODAN for Targeted Searches (Dorking) Performing an ASN Search

- The **asn** filter in SHODAN enables users to search for devices or services belonging to a specific Autonomous System Number (ASN)—a network or group of networks managed by a single organisation, such as an Internet Service Provider (ISP), hosting provider, or corporation.
- In this example, the following filter was used:
asn:AS14061



Analysis of SHODAN Output

- The **asn** filter used in this search enabled us to identify devices hosted within the network managed by DigitalOcean (ASN: AS14061).
- The results include typical information such as hostnames, labels, and response data. However, based on some responses, it appears that certain devices may be firewalled or responding with empty headers, indicating limited or restricted accessibility.

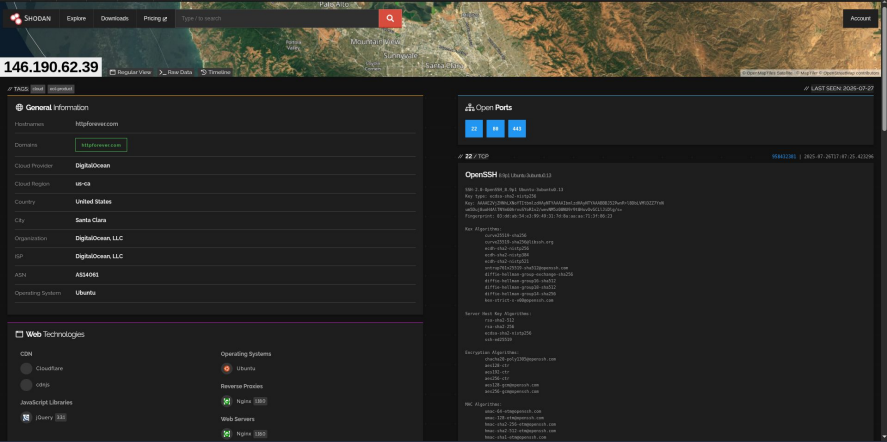
Shodan Scan Of HTTP Site - <httpforever.com>

We are going to start by scanning the first victim <httpforever.com> with Shodan

HTTP Scan (<httpforever.com>)

Step 1 - Search by domain name

- We are going to start by searching the domain name using the hostname filter
- hostname: <httpforever.com>
- We would get the following output

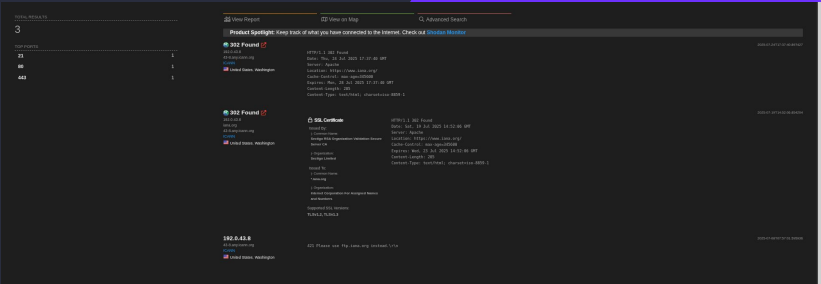


Step 2 - Understanding the fine details of the scan

- We can see that the device uses a public IP of 146.190.62.39
- We can see that the Geo Location of <httpforever.com> is in Santa Clara, US
- SSL Certificate was issued by Let's Encrypt (R3) and that they have a valid TLS certificate but expired in 2023 and so the browsers will show warnings
- We can also see 301 redirect which suggests a permanent redirect
- The server is running nginx 1.18.0 on Ubuntu where it is redirecting from HTTPS to HTTP and suggests that this version is old and probably has some known CVEs
- We can also see that the response is a HTML web page which maintains an ongoing connection that can accept multiple requests without needing to restart the session with the server
- The provider is Digital Ocean where the cloud region is us-ca
- The ASN is AS14061
- We can see that it has open ports like 22, 80, 443
- This means potentially that this site is susceptible to brute force attacks

Step 3 - Attack Plans

- Given the information we could potentially do the following:
 1. Reconnaissance by identifying the open port
 2. Check for CVEs which affect either nginx 1.18.0 or OpenSSH 8.9p1
 3. Verify HTTPS behaviour by curling the site



Shodan Scan Of HTTPS Site - iana.org

We are going to start by scanning the second victim iana.org with Shodan

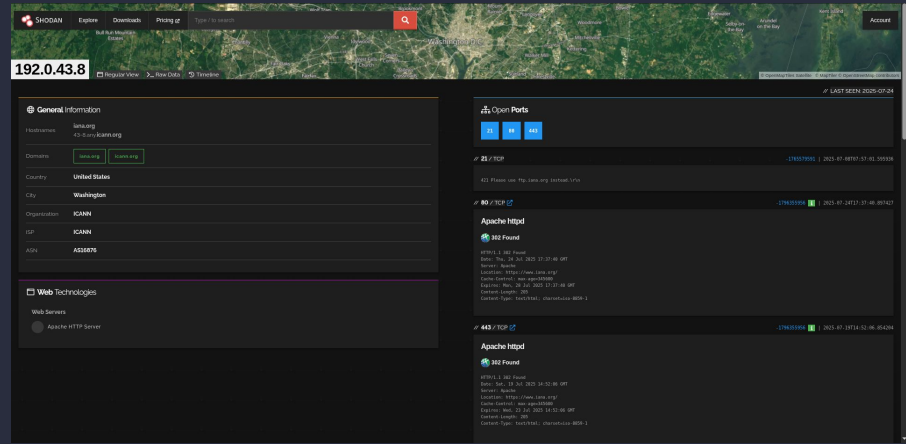
HTTPS Scan (iana.org)

Step 1 - Search by IP

- We are going to start by searching the domain name using the net filter
- First we would need to determine the IP and we would do this with nslookup in the following way:

```
tiang@LMD10:~$ nslookup iana.org
Server:      192.168.100.1
Address:     192.168.100.1#53
Non-authoritative answer:
Name:   iana.org
Address: 192.0.43.8
Name:   iana.org
Address: 2001:500:88:200::8
```

- Then we would apply the net filter for iana.org at 192.0.43.8
- We would get the following output

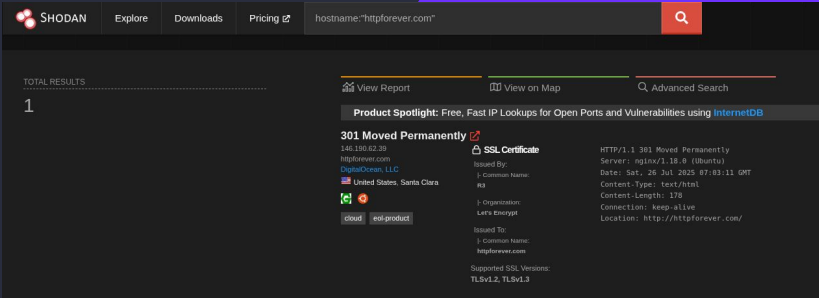


Step 2 - Understanding the fine details of the scan

- The IP clearly maps to ICANN/IANA, a globally recognised and heavily secured organisation
- This alone tells an attacker this is a high-value target with likely strong security measures and monitoring
- Ports 21, 80, 443 are open
- Knowing the exact web server allows attackers to tailor exploits if a vulnerability is known for that Apache version
- However, this IP uses a well maintained, standard web server - less likely to have outdated vulnerable software
- Has valid, recent RSA 4096-bit certification from a trusted CA
- This would indicate HTTPS is enforced, protecting against man-in-the-middle (MITM) attacks
- Certificate transparency logs are present, showing a commitment to public security standards
- HTTP/HTTPS both redirect to the canonical domain URL meaning attackers can't easily access or fingerprint the server by IP directly
- It also means not direct IP-based web content, so it is harder to exploit or fingerprint behind the scenes
- IP belongs to ICANN in Washington so attackers would know it is a critical internet infrastructure
- Overall this is much more hardened than httpforever.com

Step 3 - Attack Plan

- Could identify open ports and services to scan for known vulnerabilities on Apache HTTP Server versions or SSL implementations
- Might look for SSL/TLS vulnerabilities or weak configurations
- Phishing or social engineering
- Potentially DDoS attacks



Conclusion

Through this project, I gained valuable insights into how public internet scanning tools like Shodan can be used to analyze and compare the exposure, configuration, and security posture of different web servers.

By comparing `httpforever.com`, a basic HTTP website, to `iana.org`, a critical internet infrastructure domain managed by ICANN, the differences in network security became immediately clear.

- `httpforever.com` had minimal configuration, with redirection behavior and less secure implementation. It exposed fewer protective measures, and in some cases, returned limited or no useful server information, which could suggest default configurations or weak management.
- `iana.org`, on the other hand, was configured with strict HTTPS enforcement, issued a valid and modern SSL certificate, and used minimal open services. It clearly showed signs of being hardened against attacks, redirecting all traffic through secure channels and offering minimal attack surface.

From a security standpoint, the comparison highlighted how:

- Misconfigurations or lack of HTTPS enforcement can lead to vulnerabilities.
- Proper SSL certificate management, limited open ports, and organization-level transparency (e.g., ASN ownership) all contribute to a secure server profile.
- Attackers rely heavily on publicly available metadata — like web server type, SSL configuration, and open ports — to plan their approach. Understanding what's revealed through Shodan is the first step in reducing that exposure.

In conclusion, this project demonstrated how useful passive reconnaissance tools are — both for attackers in the early stages of an attack and for defenders looking to audit and improve their infrastructure. It also emphasized the importance of configuring web services securely and being aware of what information is exposed to the internet at large.